WORKSHEET

# Chapter 1: Points in 3D space

**1** Describe the location of each point in relation to the origin using the terms 'forward', 'back', 'up', 'down', 'left' and 'right'.

**a** $(2, 0, 4)$

**b** $(1, -5, 2)$

**c** $(-3, 1, 2)$

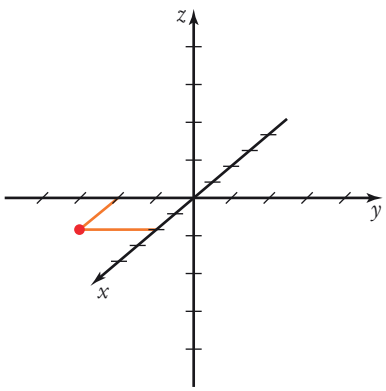
**d** $(-2, 6, -4)$

**e** $(2, -4, 1)$

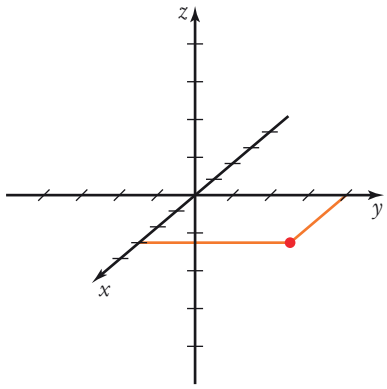
**f** $(0, -5, -3)$

**2** Write the coordinates of each point.

**a**

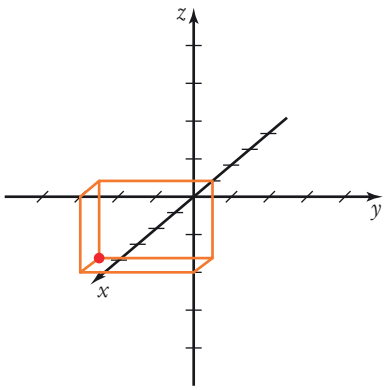



**b**

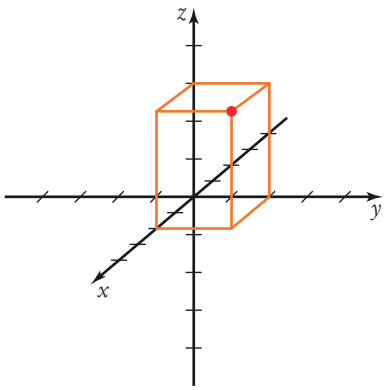



© Cengage Learning Australia Pty Ltd 2014

**c**

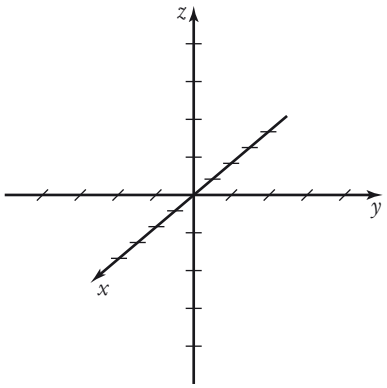



**d**

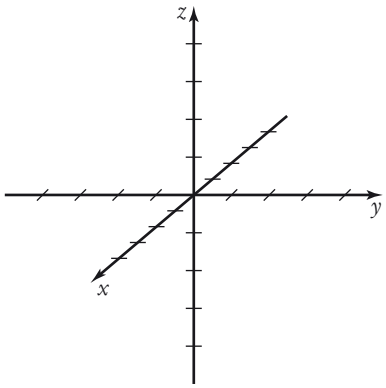



**3** Plot each point on the axes provided:

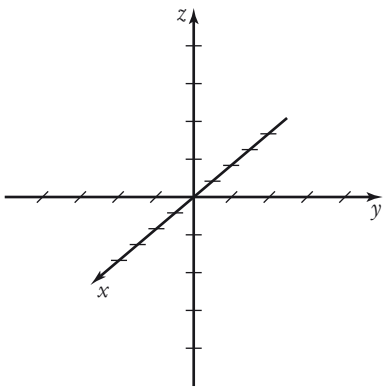
**a** $(1, -1, 0)$



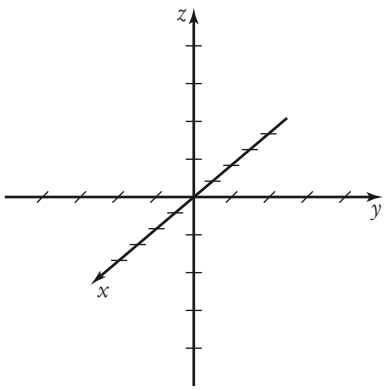

**b** $(3, -4, 2)$




**c** $(-1, 2, 2)$

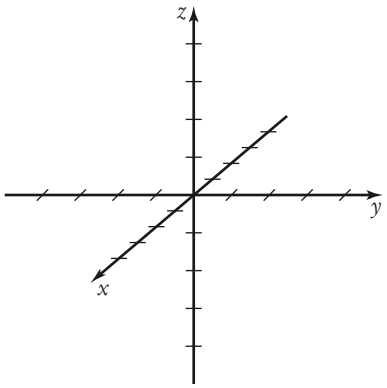



**d** $(-2, -4, 1)$

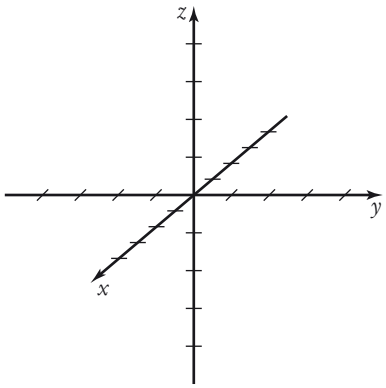



© Cengage Learning Australia Pty Ltd 2014

**e** $(-4, 3, 1)$




**f** $(-4, -3, 3)$




**4** $P(2, -3, 1)$ and which of the following points are on a line parallel with the $y$-axis?

**A** $(2, -3, -1)$

**B** $(4, 3, -1)$

**C** $(2, 3, -1)$

**D** $(2, 3, 4)$

**E** $(2, 4, 1)$

**5** $P(4, 2, -5)$ and which of the following points are on a line parallel with the $z$-axis?

**A** $(4, -2, 1)$

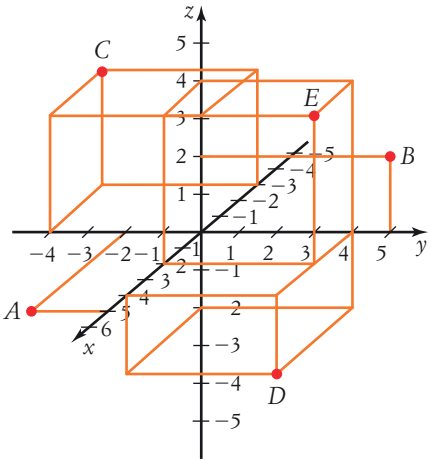
**B** $(4, 2, 0)$

**C** $(-4, 2, 5)$

**D** $(-4, 2, -3)$

**E** $(3, -2, -5)$

Use the diagram below to answer Questions 6–9.



**6** The point lying on the *yz*-plane is:

**A** $A$

**B** $B$

**C** $C$

**D** $D$

**E** $E$

**7** Point $A$ is best described as lying on the:

**A** $xz$-plane

**B** $x$-plane

**C** $xy$-plane

**D** $z$-plane

**E** $yz$-plane

**8** Name any two points that are coplanar.

**9** State the coordinates of $A$, $B$, $C$, $D$ and $E$.

**10** If $A$ and $B$ are opposite vertices of a rectangular prism with faces parallel to the $xy$-, $xz$- and $yz$-planes, find the coordinates of the other vertices.

**a** $A(0, 0, 0)$, $B(-1, 3, 4)$

**b** $A(2, -3, 4)$, $B(-1, 2, 0)$

© Cengage Learning Australia Pty Ltd 2014

## Answers

**1** **a** 2 forward, 4 up

**b** 1 forward, 5 left, 2 up

**c** 3 back, 1 right, 2 up

**d** 2 back, 6 right, 4 down

**e** 2 forward, 4 left, 1 up

**f** 5 left, 3 down

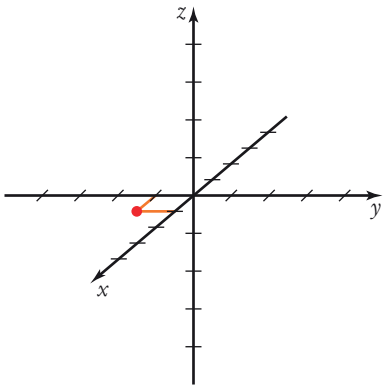
**2** **a** $(2, -2, 0)$

**b** $(3, 4, 0)$

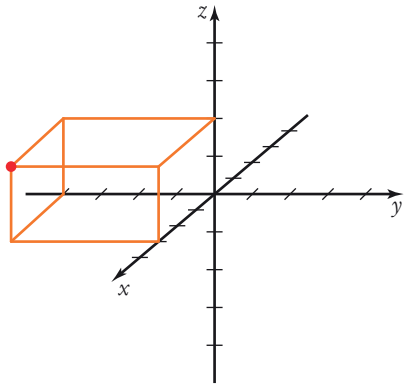
**c** $(-1, -3, -2)$

**d** $(2, 2, 3)$

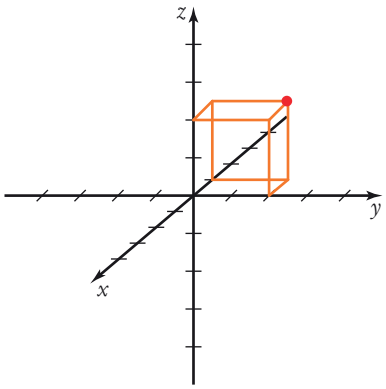
**3** **a**



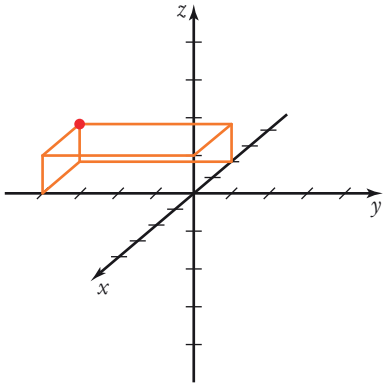

**b**




**c**

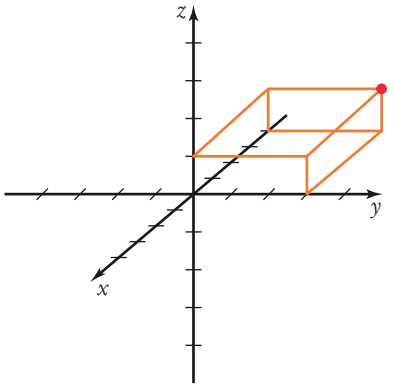



**d**

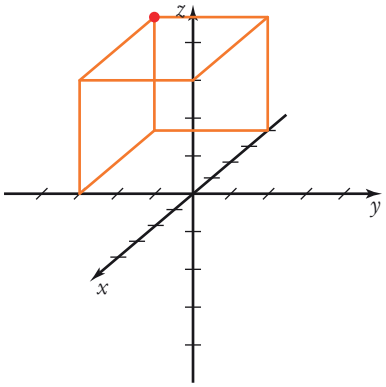



© Cengage Learning Australia Pty Ltd 2014

**e**




**f**




**4** E

**5** B

**6** B

**7** C

**8** $C$ and $E$ or $D$ and $E$

**9** $A\ (5, -2, 0)$, $B\ (0, 5, 2)$, $C\ (-3, -4, 3)$, $D\ (4, 4, -2)$, $E\ (2, 4, 4)$

**10 a** $(-1, 0, 0)$, $(0, 3, 0)$, $(0, 3, 4)$, $(-1, 3, 0)$, $(-1, 0, 4)$, $(0, 0, 4)$

**b** $(2, -3, 0)$, $(2, 2, 0)$, $(-1, -3, 0)$, $(-1, -3, 4)$, $(-1, 2, 4)$, $(2, 2, 4)$

© Cengage Learning Australia Pty Ltd 2014